Springer–Borel images and generalized Cayley maps

Integral structures, geometric fibers,
and extensions of a representation-theoretic program

Research program of Sean Rogers
A detailed statement for mathematical collaborators

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This statement connects the published work of Rogers, Kumar–Rogers, and Kumar–Xie with subsequent working results and research directions. It uses the notation of the Kumar papers wherever possible. The completed working results are *unpublished repository deductions*: internal audits and exact calculations support them, but do not constitute external peer review or establish priority. A theorem below carries that status unless expressly identified as published.

Proof-note baseline: `9f6c000`. Historical expositions are preserved; their superseded open-case counts do not govern the status reported here. The document is an exposition and research statement, not a replacement for the complete proofs or a journal submission.

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1 The central question and the shape of the program

The starting point is the relationship between representations and Schubert calculus. For a general linear group, polynomial representations and Grassmannian cohomology are related by an explicit ring map: a Schur character maps to its corresponding Schubert class when the partition fits the Grassmannian rectangle, and to zero otherwise. This gives a concrete connection between tensor products and intersection products. The Springer–Borel construction asks how to retain such a connection for other reductive groups, where there is no equally immediate notion of a polynomial character.

The published answer uses a representation-dependent Springer morphism to transport polynomials on a Cartan subalgebra into characters on a maximal torus. Taking Levi invariants and applying the Borel characteristic map gives a homomorphism to flag cohomology. The present program studies the information hidden by working only over the complex numbers: integral source lattices, divisibility of image classes, finite-rank relations, and the behavior of these images in other cohomology theories. A second branch studies the geometry of the same morphism before the Borel quotient: its degree, differential, regularity, and fibers.

These branches have a common input but different outputs. Integral-image questions ask which cohomology classes can be represented by specified polynomial characters. Cayley-map questions ask what the representation-dependent map does to points and schemes. A calculation of a generic degree does not determine a Smith normal form; a classification of integral source lattices does not determine the singularities of a fiber. A useful research statement must keep both the connection and the distinction visible.

1.1 Three successive problems

For a representation ρ and a parabolic P with Levi L, the arithmetic part has three stages:

$$\rho \quad \longmapsto \quad \Lambda_\rho \quad \longmapsto \quad A_{\rho,L} = \mathrm{Sym}_{\mathbb{Z}}(\Lambda_\rho)^{W_L} \quad \longmapsto \quad J_{\rho,L} = \beta^P(A_{\rho,L}).$$

The first stage is now controlled by a general theorem, with a closed formula for irreducible representations in every finite simple type. The second involves integral invariant theory: even if the torus source is a symmetric algebra, its Levi invariants can need higher generators and integral relations. The third is an embedded-lattice problem in integral Schubert cohomology. Neither the rational rank nor an abstract presentation of the source settles it.

The most developed ordinary results are complete Smith-factor formulas for the specified normalized single-tail images in types B and D, a primitive type C reference image, and complete relative root/weight comparisons in E6 and E7. The newest work makes the classical formulas constructive: adapted bases identify individual divisible classes, and geometric restriction kernels organize all their denominators. A short generating theorem is proved when the smaller block has size two. Its higher-block extension is the immediate structural problem.

1.2 What this statement claims

The principal current claims are stated with their hypotheses below. “Complete” always refers to a specified object: for example, all factors of a selected normalized image, or all relative root/weight factors, rather than every integral image associated with a group. Exact finite computations can establish a finite calculation; they corroborate but do not establish an arbitrary-rank theorem. Where

a proof is computer-assisted, its algebraic certificate and its geometric interpretation are separate dependencies.

The research opportunity is to turn explicit maps into structural information. The intended progression is from a character formula to an integral lattice, from a lattice to a filtration, and from a filtration to a geometric or algebraic explanation. Further examples are useful when they test one of these transitions. They are not the principal objective.

Status and sources. This overview synthesizes the research ledger and proof-status record at the pinned baseline. It introduces no additional theorem. Published foundations are distinguished in the following sections.

2 Notation, flag geometry, and the two kinds of coordinates

Let G be a connected complex reductive algebraic group, $T \subset B \subset G$ a maximal torus and Borel subgroup, and $P \supseteq B$ a standard parabolic with Levi subgroup $L \supset T$. Write

$$X = X^*(T), \qquad Q = \mathbb{Z}\Phi \subset X, \qquad W = N_G(T)/T, \qquad W_L = N_L(T)/T.$$

Here Φ is the root system and Q the root lattice. For a simply connected simple group, X is the full weight lattice. We use P for a parabolic, rather than also using it for the weight lattice. A simple root is α_i, its coroot is $\alpha_i^\vee$, and the fundamental weights satisfy $\omega_j(\alpha_i^\vee) = \delta_{ij}$.

A weight μ has two uses. The symbol e^μ denotes its multiplicative torus character, an element of $\mathbb{Z}[X]$. Its differential, also denoted μ, is an additive linear coordinate on $\mathfrak{t} = \mathrm{Lie}(T)$. Thus $e^{\mu+\nu} = e^\mu e^\nu$ in the character ring, while $\mu + \nu$ is addition of linear functions in $\mathrm{Sym}(X)$. Their relationship through the Springer map is not the exponential map.

2.1 Parabolics and classical parameters

A retained set I of simple nodes generates the Levi Weyl group W_I. Omitting one node gives a maximal parabolic; omitting several gives a partial flag with several steps. The empty retained set gives G/B, and the full retained set gives a point. Consequently an all-parabolic statement in rank r has $2^r - 1$ proper standard cases, including the full flag.

For the families in Parts II and III, write $a = n - k$. The varieties and matrix Levi patterns are

type	G/P_{n-k}	matrix Levi pattern
C_n	$\mathrm{IG}(a, 2n)$	$\mathrm{GL}(a) \times \mathrm{Sp}(2k)$
B_n	$\mathrm{OG}(a, 2n+1)$	$\mathrm{GL}(a) \times \mathrm{SO}(2k+1)$
D_n	$\mathrm{OG}(a, 2n)$	$\mathrm{GL}(a) \times \mathrm{SO}(2k)$.

The parameter n is the ambient rank, a the isotropic subspace dimension, and k the residual classical rank. The type D results below use $a \geq 1$, $k \geq 2$, and the appropriate simple-rank range. Maximal isotropic components require their own conventions. When the group is Spin, the Levi is the inverse image of the matrix Levi under the central isogeny; it must not be identified with the displayed product without retaining that cover.

This distinction explains why the published work uses orthogonal groups even though later uniform theorems are formulated for simply connected simple groups. The flag varieties can agree under a central isogeny, while the allowed characters and integral source lattices differ. Lie type alone does not specify an integral representation-ring problem.

2.2 Bruhat decomposition and integral coordinates

Let W^P be the minimal representatives for W/W_L. The Bruhat cells are affine spaces indexed by W^P; their closures and the opposite Schubert varieties give an integral basis

$$\epsilon_w = [\overline{B^- wP/P}] \in H^{2\ell(w)}(G/P;\mathbb{Z}).$$

The cell decomposition implies that this cohomology is torsion-free and concentrated in even degrees. Thus torsion in an image quotient is arithmetic of the chosen image, not torsion in the ambient cohomology. For a Grassmannian $\mathrm{Gr}(\ell,n)$, the same basis is indexed by partitions $\lambda \subseteq (n-\ell)^\ell$.

We use polynomial degree d for cohomological degree $2d$. A second, explicitly declared grading occurs for square generators: a variable y_i representing x_i^2 has *source weight* one but polynomial degree two, hence cohomological degree four. In the K-theory specialization formulas, a grading is transported from a universal homogeneous model; it is not an intrinsic grading of the K-ring. Keeping these conventions separate prevents apparent disagreements between formulas.

The Borel presentation over $\mathbb{Q}$ is

$$H^{2*}(G/P;\mathbb{Q}) \simeq \mathrm{Sym}_{\mathbb{Q}}(X_{\mathbb{Q}})^{W_L}/(\mathrm{Sym}_{\mathbb{Q}}(X_{\mathbb{Q}})^W_+).$$

The ideal is generated in the Levi-invariant ring by the indicated positive-degree global invariants. Over $\mathbb{Z}$, this presentation cannot be used without additional hypotheses. Divided differences $\partial_i f = (f - s_i f)/\alpha_i$ and integral Schubert coordinates supply an exact alternative. The Chevalley formula gives a second way to compute divisor products. Comparing these two routes is particularly useful at the primes at which invariant-ring shortcuts fail.

Status and sources. Bruhat decomposition, Schubert bases, characteristic maps, and divided differences are published foundations; see Brion [6] and the sources collected in the mathematical framework. Their use here is background, not a new contribution.

3 The published Springer–Borel construction and its extensions

Let $\rho : G \to \mathrm{GL}(V)$ have faithful differential and nondegenerate trace form

$$B_\rho(X,Y) = \mathrm{tr}(d\rho(X)d\rho(Y)).$$

For a simple complex group and a nontrivial actual representation, this form is nondegenerate. Trace-orthogonal projection defines a conjugation-equivariant regular map $\theta_\rho : G \to \mathfrak{g}$ by

$$B_\rho(\theta_\rho(g), X) = \mathrm{tr}(\rho(g)d\rho(X)).$$

In the semisimple normalization, $\theta_\rho(1) = 0$ and $d\theta_\rho|_1 = \mathrm{id}$. Write $\Theta_\rho = \theta_\rho|_T : T \to \mathfrak{t}$.

3.1 Explicit determination

If the weights are columns $\mu = (\mu(\alpha_i^\vee))_i$ with multiplicities m_μ, put

$$M_\rho = \sum_\mu m_\mu \mu\mu^t, \qquad v_\rho(t) = \sum_\mu m_\mu e^\mu(t)\mu.$$

In the simple-coroot basis, the coefficient column of $\Theta_\rho(t)$ is $M_\rho^{-1}v_\rho(t)$. This is Rogers's explicit weight-moment method [2]. Its practical consequence is that the map can be reconstructed from weights and multiplicities, independently of a printed coordinate table. The repository copy consulted is the 2017 preprint; the final journal version was not available for a line-by-line comparison.

3.2 The map of Kumar

The polynomial-character ring and characteristic homomorphism are

$$\mathrm{Rep}^{\mathbb{C}}_{\rho\text{-poly}}(L) = \Theta^*_\rho(\mathrm{Sym}_{\mathbb{C}}(\mathfrak{t}^*)^{W_L}), \qquad \xi^P_\rho = \beta^P_{\mathbb{C}} \circ (\Theta^*_\rho)^{-1}.$$

Part I constructs this map and proves surjectivity for the full complex Levi polynomial source [1]. The inverse is defined on the image of an injective pullback. It is not a rational inverse of $T \to \mathfrak{t}$. Our integral questions refine this construction by specifying which transported polynomials pull back to integral virtual characters.

Part II studies the residual symplectic and odd orthogonal sources and proves stable injectivity as $n \to \infty$ with k fixed [3]. Part III supplies the even orthogonal counterpart and exceptional maximal-parabolic computations, excluding E8 [4]. In the papers' tail coordinates $\bar{h}_j = (h_j - h_j^{-1})/2$, the square generators are $e_i(\bar{h}_1^2, \ldots, \bar{h}_k^2)$; type D also has $p = \prod \bar{h}_j$, with $p^2 = e_k$. The published Chern-class expression for a square generator is

$$e_i \longmapsto c_i^2 + 2\sum_{j=1}^{i}(-1)^j c_{i+j}c_{i-j}, \qquad c_0 = 1,$$

with the quotient-bundle conventions of those papers. Stable injectivity does not assert finite-rank injectivity or integral saturation.

3.3 What the subsequent work adds

The earlier combinatorial proof project supplies alternative finite and stable arguments in A, C, and D, including the Euler generator. It does not claim that the published main theorems lacked proofs. The later integral work asks a different question: after selecting a $\mathbb{Z}$-form, how far is its finite image from the full integral lattice in its rational span? The answer requires information erased by complexification.

The exceptional-table review independently regenerated the weights and matrix inverses for F4, E6, and E7. It agreed with the F4 and E7 torus formulas and proposed replacing the six denominators 6 by 18 in Part III's E6 Lemma 7.1, with unchanged numerators. Additional Schubert-table corrections are recorded separately. These remain proposed corrections supported by reproduction files, not a published erratum. The purpose of including this history is provenance: the present calculations descend from the published weight method, while their correctness is tested without importing the original tables.

The Sage library grew from the original evaluation programs into general character, BGG, Schubert-product, inverse-Borel, lattice-comparison, and exceptional-map routines. "Inverse Borel" means finding polynomial representatives for specified cohomology classes when they lie in the chosen image; it does not invert the Borel map uniquely. Reducing in the coinvariant quotient and reusing Weyl data made larger computations feasible. These are mathematical infrastructure and reproducibility improvements; their existence alone is not a claim of a new theorem.

Status and sources. Published formulas and stable results are attributed above. Alternative proofs, corrections, and implementation are indexed in the article collection. Each correction should be reviewed against its stated source version before being circulated as an erratum.

4 A general integral source theorem

Assume now that G is simple and simply connected and ρ has faithful differential. Define

$$\mathcal{C}_\rho = \{f \in \mathrm{Sym}_{\mathbb{Q}}(X_{\mathbb{Q}}) : \Theta_\rho^* f \in \mathbb{Z}[X]\}, \qquad \Lambda_\rho = \mathcal{C}_\rho \cap X_{\mathbb{Q}}.$$

Integral virtual characters are intended, so negative coefficients are allowed. Integrality is a condition on the character, not on its expression in a chosen rational coordinate list.

Theorem 4.1 (Full torus source; repository deduction)**.** *The lattice Λ_ρ has finite index in X, and*

$$\mathcal{C}_\rho = \mathrm{Sym}_{\mathbb{Z}}(\Lambda_\rho), \qquad A_{\rho,L} = \mathrm{Sym}_{\mathbb{Z}}(\Lambda_\rho)^{W_L}.$$

These equalities include inhomogeneous polynomials. The Borel image of $A_{\rho,L}$ lies in $H^{2*}(G/P;\mathbb{Z})$.

Here is the central integral argument. Choose a basis of the full linear lattice and write its Laurent pullbacks as columns of an integer coefficient matrix C. By definition these columns form a primitive sublattice: an integral Laurent polynomial in their rational span already comes from a lattice element. Thus C has full column rank modulo every prime. Row reduction modulo a prime produces Laurent polynomials with independent leading exponents. The weight-moment formula shows that a nonzero coefficient minor corresponds to linearly independent exponent vectors over $\mathbb{Q}$. Distinct monomials in the reduced generators therefore have distinct leading terms. They are algebraically independent in every residue characteristic, even when an exponent determinant is divisible by that characteristic.

If a rational polynomial in these generators had an integral pullback and a nontrivial denominator, reduction at a prime dividing a least denominator would contradict that independence. This proves the symmetric-algebra assertion. Differentiating an integral Laurent pullback at the identity gives an integral weight; since $d\Theta_1 = \mathrm{id}$, this also proves $\Lambda_\rho \subset X$. Finally, full-flag characteristic integrality and primitive Schubert pullback from G/P prove the parabolic assertion. No averaging by $|W_L|$ is used.

4.1 Closed formula for irreducibles

Normalize the invariant inner product so long roots have squared length two. For a nonzero dominant weight λ, let ϱ be the half-sum of positive roots and define

$$I_\lambda = \frac{\dim V(\lambda)}{\dim \mathfrak{g}}(\lambda, \lambda + 2\varrho), \qquad D_\lambda = \sum_\mu \mathbb{Z}(m_\mu \mu).$$

This is the Dynkin-index convention with $I_{\mathrm{Ad}} = 2h^\vee$; see [8]. For a lattice M, put $M^* = \{v : (v, M) \subseteq \mathbb{Z}\}$.

Theorem 4.2 (Multiplicity-weighted lattice; repository deduction)**.** *For every nontrivial irreducible representation of every finite simple type,*

$$D_\lambda = Q + \mathbb{Z}\lambda, \qquad \Lambda_\lambda = I_\lambda (Q + \mathbb{Z}\lambda)^*.$$

The nontrivial point is the multiplicity-weighted equality, not the usual span of all weights. Along a shortest Dynkin path from the support of λ to any chosen simple node, successive lowering steps give multiplicity-one weights. The path is forced because the finite Dynkin diagram is a tree. Differences of consecutive weights yield each simple root, while the highest weight itself has multiplicity one.

This proves $Q+\mathbb{Z}\lambda \subseteq D_\lambda$; the reverse inclusion follows from the common weight coset. Trace duality then gives Λ_λ.

For E6 and E7 this becomes $I_\lambda X$ for trivial central character and $I_\lambda Q$ otherwise. For E8 it is $I_\lambda X = I_\lambda Q$; for G2 and F4 it is $I_\lambda Q_{\mathrm{long}}$ under the specified metric identification. The theorem eliminates a separate weight enumeration for the linear source of each irreducible. It does not eliminate integral invariant theory after taking W_L invariants. Nor does the closed irreducible formula automatically classify direct sums: multiplicities at overlapping weights combine before the coefficient lattice is formed. Repeated copies of one representation leave the normalized map unchanged.

Status and sources. Full proofs and internal literature audits: torus source, source audit, irreducible classification, and classification audit. These are general repository deductions from published trace and highest-weight theory, with priority unresolved.

5 Integral images: the objects and the general bound

Suppose $J_d \subset H^{2d}(G/P;\mathbb{Z})$ is the image of a specified integral source in degree d. Its saturation is

$$S_d = \mathrm{Sat}(J_d) = (J_d \otimes \mathbb{Q}) \cap H^{2d}(G/P;\mathbb{Z}).$$

There are integral bases in which $J_d \hookrightarrow S_d$ is diagonal with positive entries $s_1 \mid \cdots \mid s_r$. These are its Smith factors, and $S_d/J_d \simeq \bigoplus_i \mathbb{Z}/s_i$. A unit factor records a primitive direction. If J_d has smaller rational rank than the ambient group, H^{2d}/J_d also has a free summand. That free summand is not counted by the nonzero Smith factors.

For example, $\mathbb{Z}[2h] \subset \mathbb{Z}[h]/(h^{N+1})$ has factor 2^d in degree $2d$. Over $\mathbb{Q}$ the image is the whole ring; over $\mathbb{Z}$ it misses primitive classes. This example already shows why a complex surjectivity theorem cannot answer the integral question.

5.1 Literal and normalized sources

There are two legitimate but different arithmetic problems. The *literal* source consists of polynomials whose Springer pullbacks are integral virtual characters. The *normalized* classical source uses the convenient generators $e_i(\bar{h}^2)$ and, in D, $\prod \bar{h}_j$ from the papers. Since $\bar{h} = (h - h^{-1})/2$, these need not themselves be integral characters. For the specified SO/Sp tail forms, the literal generators are $4^i e_i$ and $2^k p$. Thus in polynomial degree d the literal image is 2^d times the normalized image. This statement has a specified character lattice and does not automatically describe every Spin source.

Theorem 5.1 (Prime-local full-source bound; repository deduction)**.** *For simple simply connected G, let t_G be its torsion index for X, let $e_\rho = \exp(X/\Lambda_\rho)$, and let $J^d_{\rho,L}$ be the full literal image. Then*

$$t_G e_\rho^d H^{2d}(G/P;\mathbb{Z}) \subseteq J^d_{\rho,L} \subseteq H^{2d}(G/P;\mathbb{Z}).$$

In particular the image is the whole group after localization at any prime not dividing $t_G e_\rho$.

The torsion-index characteristic-map argument is established machinery. The additional source inclusion is $e_\rho X \subseteq \Lambda_\rho$. Applying divided differences to a torsion-index top representative supplies the parabolic invariant representatives without a spurious factor $|W_L|$. The theorem bounds all bad-prime exponents but rarely determines them sharply. It applies to the full literal Levi source, not to every selected smaller tail algebra.

An exact type A benchmark is the defining representation of SL_m on $\mathbb{P}^{m-1}$. The full linear source is Q, the root lattice. With h the positive hyperplane class, the full literal degree-d image is

$$J_H^d = \frac{m}{\gcd(m,d)}\mathbb{Z}h^d \quad (1 \le d < m), \qquad J_H^0 = \mathbb{Z}.$$

The Levi-invariant elementary generators map to $\binom{m}{j}h^j$. Their products have precisely the displayed gcd content. Even this familiar variety therefore has nontrivial representation-dependent arithmetic. The formula concerns the full root-polynomial source; substituting a single generator would change the problem.

Status and sources. See the bound and projective calculation, its audit, and normalization conventions. Torsion-index and characteristic-map inputs are published; compare [15]. The sharp source-specific formulas are repository deductions.

6 Complete classical Smith formulas

6.1 Type C as the primitive reference

For the normalized residual symplectic square generators on $\mathrm{IG}(a,2n)$, the selected image is primitive in its rational span. Its nonzero Smith factors are all one. This is the type C counterpart of the B/D calculation, but its arithmetic answer is simpler. Quaternionic Grassmannian geometry and integral Schur classes give the reference lattice. The B/C comparison then records the powers of two introduced by the change in root lengths and Schubert normalization.

6.2 Type B in every rank

Set $n = a + k$, $\ell = \min(a,k)$, $A = \max(a,k)$, $r = \lfloor n/2 \rfloor$, and $c = \lceil n/2 \rceil$. Use the normalized tail generators $q_i = e_i(x_{a+1}^2, \dots, x_n^2)$ on $\mathrm{OG}(a, 2n+1)$. Let $\begin{bmatrix} v \\ u \end{bmatrix}_q$ denote a Gaussian binomial, zero outside $0 \le u \le v$, and put

$$U_{N,L}(q) = \sum_{j=0}^{\lfloor L/2 \rfloor} q^{j(\lceil N/2 \rceil - L + j)} \begin{bmatrix} \lfloor N/2 \rfloor \\ j \end{bmatrix}_q \begin{bmatrix} \lceil N/2 \rceil - j \\ L - 2j \end{bmatrix}_q, \qquad U_{0,0} = 1.$$

Theorem 6.1 (Normalized B multiplicities; repository deduction)**.** *If $M_{n,\ell,e}(m)$ counts the factor 2^e in source weight m, then*

$$\sum_m M_{n,\ell,e}(m) q^m = q^{ec} \begin{bmatrix} r \\ e \end{bmatrix}_q U_{n-2e,\ell-e}(q), \qquad 0 \le e \le \ell.$$

Source weight m means polynomial degree $2m$ and cohomological degree $4m$. There are no image classes in odd polynomial degree.

This is a formula for all ranks and degrees in the stated single-tail problem. The positive Gaussian expression has earlier counting interpretations; the additional assertion is its identification with integral Springer-image multiplicities. It does not assign a Smith exponent directly to every original Schur generator.

The proof starts with a type C coefficient-row lattice M_m and a filtration $P_t(m)$ generated by rows of defect at most t. The B row lattice is $\sum_t 2^t P_t(m)$. To infer Smith factors one must prove

each prefix primitive over $\mathbb{Z}$. A rank count over $\mathbb{Q}$ would not suffice. The integral proof combines Schur-product rows, a geometric upper bound from a square-zero incidence problem, and a triangular unimodular minor. Once every prefix is primitive, a common adapted basis diagonalizes the weighted filtration. Its rank jumps are the displayed multiplicities.

6.3 Type D and Euler gluing

The type D source contains both squares and the Euler generator $p = x_{a+1} \cdots x_n$, with $p^2 = q_k$. Its square-sector row lattice agrees with the corresponding B row lattice, not merely with its Smith list. The Euler sector is related to a shifted smaller-block problem. The remaining issue is the integral position of the two rationally independent sectors.

Define

$$B_{a,k,e}(q) = q^{e\lceil (a+k)/2 \rceil} \begin{bmatrix} \lfloor (a+k)/2 \rfloor \\ e \end{bmatrix}_q U_{a+k-2e,\min(a,k)-e}(q).$$

Outside its allowed exponent range this is zero; set $B_{0,k,0} = 1$. For even $k = 2h$ put

$$I_{a,2h}(q) = q^{h(\lfloor a/2 \rfloor + 1)} \begin{bmatrix} h + \lceil a/2 \rceil - 1 \\ h \end{bmatrix}_q,$$

and put $I_{a,k} = 0$ for odd k.

Theorem 6.2 (Full normalized D multiplicities; repository deduction)**.** *For $a \geq 1$, $k \geq 2$, if $M_D(a, k; e, d)$ counts factors 2^e in polynomial degree d, then*

$$\sum_d M_D(a, k; e, d) z^d = B_{a,k,e}(z^2) + z^k B_{a-1,k,e}(z^2) + (\delta_{e,1} - \delta_{e,0}) I_{a,k}(z^2).$$

All gluing takes place at unit levels of the two sector lattices; there is no additional positive-level gluing.

The correction converts the appropriate unit factors into factors two. Omitting this step would leave the full D theorem incomplete even if each sector were separately understood. The theorem has now been proved in the repository; the earlier small-block cases are independent controls, not the boundary of current coverage. Literal SO vector-character factors are obtained by the stated degree scaling. Arbitrary Spin sources require their own source comparison.

Status and sources. See the all-rank B proof, its audit, the D square comparison, and the full D theorem. The general results use published integral Schubert geometry and symmetric functions. They remain unpublished deductions, not externally reviewed theorems.

7 From multiplicities to classes and multiplication

A Smith list tells us the sizes of quotient groups but not which integral cohomology classes represent them. The current ordinary work extracts that additional structure. Put the smaller block at the head and write

$$R = H^{2*}(\mathrm{Gr}(\ell, n); \mathbb{Z}) = \mathbb{Z}[e_1, \ldots, e_\ell]/(h_{A+1}, \ldots, h_{A+\ell}), \quad n = A + \ell.$$

Here e_i, h_i are elementary and complete symmetric functions in variables $y_1, \ldots, y_\ell$ of weight one. The map ϕ sends these symmetric polynomials to the normalized square image, so weights are doubled in polynomial degree. After the finite kernel has been taken, R embeds through ϕ into its rational image.

7.1 A simultaneous basis

The primitive-prefix proof provides one unimodular row basis V adapted to every P_t. If a row enters at level e_γ, the coordinate-dual basis element f_γ, given by the corresponding column of V^{-1} in Schur coordinates, yields

$$b_\gamma = \frac{\phi(f_\gamma)}{2^{e_\gamma}}.$$

These are primitive integral classes in the saturation. "Dual" here means dual to coefficient coordinates; it is not the Poincare pairing. This distinction matters when transporting products.

If $f_\alpha f_\beta = \sum_\gamma C^\gamma_{\alpha\beta} f_\gamma$ in R, then

$$b_\alpha b_\beta = \sum_\gamma 2^{e_\gamma - e_\alpha - e_\beta} C^\gamma_{\alpha\beta} b_\gamma.$$

The coefficients are integers because the saturation of a rational subalgebra in torsion-free integral cohomology is a ring. This gives an exact multiplication algorithm with a proof of integrality, rather than a Smith diagonalization disconnected from the product. The square-sector B/D row equality transports this saturated ring structure between the two cases.

For the full D source, start with the adapted bases in the square and Euler sectors and then perform the unit-level gluing. Binary row reduction identifies the glued unit directions; integral lifts and complementary rows complete a unimodular basis. Products must still use $p^2 = q_k$ and the correct smaller Euler rectangle. The two-sector decomposition is additive, not a product decomposition of rings.

7.2 A divided-power part of the saturation

Let $c = \lceil n/2 \rceil$. If every part of a partition ν lies between c and A and its length is at most ℓ, set

$$z_\nu = \frac{\phi(m_\nu)}{2^{\ell(\nu)}}.$$

These classes form a primitive truncated divided-power subring. Writing $m_j(\nu)$ for the multiplicity of part j, their products are

$$z_\nu z_\mu = \left(\prod_j \binom{m_j(\nu) + m_j(\mu)}{m_j(\nu)} \right) z_{\nu \cup \mu}$$

when the combined length is at most ℓ, and zero otherwise. Collisions of high powers on one root vanish because $2c \geq n$ and each root has nth power zero in the full-flag model. The intersection with the original image multiplies z_ν by $2^{\ell(\nu)}$.

This makes part of the arithmetic tangible. A factor 2^e is represented by an actual class obtained by dividing a source polynomial by 2^e, and its products obey integral divided-power coefficients. It also indicates why ordinary powers and rational differential operators can obscure the relevant arithmetic.

Status and sources. These constructions are in the adapted-basis note and the full D basis note. Separate Sage/native calculations check embeddings and products. Native Chevalley calculations supply a second route to actual Schubert coefficients; they do not replace the uniform proof.

8 Geometric denominators and the current generator conjecture

Choose a maximal-rank square-zero endomorphism N of $\mathbb{C}^n$ and define closed defect varieties

$$W_t = \{U \in \mathrm{Gr}(\ell, n) : \mathrm{rank}(N : U \to \mathbb{C}^n/U) \le t\}.$$

Thus W_0 is the locus of N-stable planes. Set $K_0 = R$ and

$$K_e = \ker\big(R \longrightarrow H^*(W_{e-1}; \mathbb{Z})\big), \quad 1 \le e \le \ell, \qquad K_{\ell+1} = 0.$$

All rings retain their integral structure. The restriction kernel concerns the particular defect variety, not an arbitrary Springer or Spaltenstein variety with a similar partition label.

Theorem 8.1 (Denominator formula; repository deduction)**.** *Under the B/D square-image identification, the complete saturation is*

$$S = \sum_{e=0}^{\ell} 2^{-e} K_e \subset R_{\mathbb{Q}}.$$

Let J_e be the ideal generated by m_ν with $\ell(\nu) \ge e$ and every part at least $c = \lceil n/2 \rceil$. Then

$$J_e \subseteq K_e \quad \textit{for every } e, \qquad J_\ell = K_\ell = (e_\ell^c).$$

The prefix theorem identifies K_e as the exact annihilator of the preceding coefficient-row prefix. This is the reason the filtration controls denominators. The inclusion for high monomials follows by pullback to an integral incidence/splitting model: with at least e high parts, not all can occupy the $e-1$ surviving residual directions, and the remaining term vanishes by root nilpotence. The top-layer equality follows by primitivity and the matching top-defect rank.

Conjecture 8.2 (Short integral generators)**.** For all $A \ge \ell$ and every e, one has $J_e = K_e$ in R.

This is a presentation conjecture, not a missing Smith-count conjecture. The abstract multiplicities and geometric denominator formula are already known within the program. Equality would express every layer through elementary symmetric-function data and give a compact generating set for the saturated image. A recurrence reduces the high parts to $c, c+1, \dots, c+\ell-1$; the candidate ring consequently has at most $\binom{2\ell}{\ell}$ module generators over R, independently of A.

8.1 The completed two-variable theorem

For $\ell = 2$, $n = A + 2$, set $u = y_1 + y_2$, $v = y_1 y_2$, and $p_j = y_1^j + y_2^j$. Then

$$R = \mathbb{Z}[u, v]/(h_{n-1}, h_n), \qquad K_1 = (p_c, p_{c+1}, v^c), \qquad K_2 = (v^c).$$

The entire saturation is

$$S = R + Rz + Rw + Rt, \qquad z = p_c/2, \quad w = p_{c+1}/2, \quad t = v^c/4.$$

The three extra homogeneous algebra generators are necessary, with weights $c, c+1, 2c$. Their products include

$$z^2 = 2t, \quad zw = ut, \quad w^2 = 2vt, \quad zt = wt = t^2 = 0.$$

Together with $2z = p_c$, $2w = p_{c+1}$, $4t = v^c$, these are useful multiplication rules; they are not being claimed as a complete defining ideal, since further R-linear relations remain.

The key proof is integral straightening. The identities

$$h_{2c-1} = p_c h_{c-1}, \quad h_{2c} = p_c h_c - v^c, \quad h_{2c-2} = p_c h_{c-2} + v^{c-1}$$

reduce the quotient by J_1 to a monic Groebner basis built from $v^j p_{c-j}$. Adjacent reductions follow from the power-sum recurrence; the boundary coefficient 2 is reduced by a monic power of v, not divided out. The resulting free integral quotient has the known defect-variety rank. Since $J_1 \subseteq K_1$ and the ranks agree, freeness proves equality over $\mathbb{Z}$.

8.2 A uniform next proof route

Let $V = \mathbb{Z}[z]/z^n$ and let T act by multiplication by z. The rectangular Schur basis identifies R, as a lattice, with $\bigwedge^\ell V$ through alternants. Put

$$A(a) = \sum_{j=c}^{n-1} a_j T^j, \qquad U(a) = \bigwedge^{\ell} (1 + A(a)).$$

Since $A(a)A(b) = 0$, coefficients of total a-degree at least e give exactly the multiplication operators by the high monomials generating J_e. The missing theorem is that their sum of integral images is primitive with the known K_e rank. This reformulation preserves divided-power coefficients and is suitable for a uniform straightening proof. Replacing the family by one generic matrix or working only over $\mathbb{Q}$ would leave the essential gap.

Status and sources. See the denominator theorem and conjecture and the two-variable proof. The remaining range is $\ell \geq 3$, $1 \leq e < \ell$. Nine exact finite ideal checks are supporting evidence only. Truncated-symmetric and current-algebra literature [10, 11, 12, 13] provides promising comparisons, not an already verified integral identification with these K_e.

9 Joint images and exceptional arithmetic

9.1 Several Levi blocks

On a symplectic two-step flag with positive block sizes a, b, k, use normalized generators $c_j = e_j(x_{\text{middle}})$ and $q_i = e_i(x_{\text{tail}}^2)$. Let L_1, L_2 be their separate images and J the algebra generated jointly. The repository deduction from integral flag geometry is that J is primitive in ambient cohomology, $L_1 + L_2$ is primitive in J and in ambient cohomology, and $L_1 \cap L_2 = \mathbb{Z}$. An integral basis simultaneously contains bases for both separate images. Mixed products contribute new free directions, not torsion in this sum quotient.

For the analogous normalized B/D flags, the sum is still primitive inside the joint image and the intersection is $\mathbb{Z}$, but ambient saturation can fail. Ambient torsion is 2-primary; its complete multiplicities remain open. These are therefore genuine extensions of the single-tail problem, rather than automatic consequences of the completed Grassmannian formulas. A multistep generalization must specify all generators and their integral positions.

9.2 Relative exceptional images

For a proper retained set I, define unscaled characteristic images

$$M_I^d = \beta^{P_I}((\mathrm{Sym}_{\mathbb{Z}}^d Q)^{W_I}), \qquad N_I^d = \beta^{P_I}((\mathrm{Sym}_{\mathbb{Z}}^d X)^{W_I}).$$

The finite quotient N_I^d/M_I^d compares two embedded full-source images. It is not the ambient quotient $H^{2d}(G/P_I;\mathbb{Z})/N_I^d$. This distinction is the principal qualification to the exceptional completeness statements.

In E6 all 63 proper standard parabolics have complete relative formulas. Sixty have a common expression. Define

$$\begin{aligned} C_1(q) &= q+q^2+q^7+q^8+q^9+2q^{10}+q^{11}+q^{12}+q^{14},\\ C_2(q) &= q^3+q^4+q^6+q^7+q^9+q^{11}+q^{12}+q^{13}+q^{15}+q^{16},\\ C_3(q) &= q^5+q^8, \end{aligned}$$

and $D(q)=[2]_q[4]_q[5]_q[6]_q[8]_q$, where $[j]_q=1+\cdots+q^{j-1}$. If I does not contain $\{1,3,5,6\}$ in Bourbaki numbering, the multiplicity of 3^a is

$$[q^d]\frac{C_a(q)D(q)}{P_{W_I}(q)}, \qquad a=1,2,3.$$

The three excluded retained sets have separate completed formulas. For the minimal 27-dimensional and adjoint 78-dimensional representations, the full literal degree-d images are $6^dM_I^d$ and $24^dN_I^d$, respectively. The scalar comparison and the nontrivial root/weight quotient are distinct parts of this answer.

In E7 all 127 proper standard parabolics satisfy

$$2N_I^d \subseteq M_I^d \subseteq N_I^d, \qquad N_I^d/M_I^d \simeq (\mathbb{Z}/2)^{m_{I,d}}.$$

For the 112 sets omitting at least one of $2,5,7$,

$$\sum_d m_{I,d}q^d = \frac{qP_{W(E7)}(q)}{(1+q)P_{W_I}(q)}.$$

The remaining fifteen have explicit completed multiplicity polynomials, including the D6, D5+A1, D4+A1 and even-block families. The first relative gap occurs in degree one for 112 sets, degree two for twelve, degree four for two, and degree eight for D6. Completeness uses full integral invariant sources, including generators missed by earlier bounded searches. A source Hilbert/leading-monoid argument proves generation in all degrees; it is not inferred from stopping at a chosen degree cutoff.

9.3 An absolute benchmark and the remaining exceptional work

For G2/B, the minimal literal source is $\mathbb{Z}[6\omega_1,2\omega_2]$. Its image has presentation

$$J_{7,T} \simeq \mathbb{Z}[X_1,Y_1]/(X_1^2-3X_1Y_1+3Y_1^2,Y_1^6).$$

In polynomial degrees 0 through 6, the Smith lists are

$$(1),\ (2,6),\ (12,12),\ (24,144),\ (288,288),\ (576,1728),\ (3456).$$

The adjoint lists are multiplied by 4^d in degree d. These are absolute ambient image factors. A monic normal-form argument proves the integral kernel, and independent BGG/Chevalley routes check the embeddings.

For F4 and E8 the root and weight lattices agree, so a nontrivial Q/X comparison cannot be the next target. Their relevant arithmetic questions concern representation scaling, full Levi invariant

sources, and ambient image cokernels. The irreducible linear-source formula is already uniform; a uniform full ambient-image classification is not. In E6/E7, a geometric explanation of the finite relative patterns remains worthwhile, but should be formulated as a new structural theorem rather than another calculation of the same table.

Status and sources. Sources: C joint theorem, B/D joint theorem, E6 transfer, E6 completion, E7 completion, and G2 full flag. All are repository deductions or finite computer-assisted results with their individual scopes; no general ambient exceptional classification is claimed.

10 K-theory: the same source, a different embedded image

The original polynomial characters already belong to a representation ring, so they have a natural K-theoretic image. Let $\kappa_P : R(L) \to K^0(G/P)$ be the associated-bundle homomorphism. For the full literal source define

$$\eta_\rho^P : A_{\rho,L} \to K^0(G/P), \qquad f \longmapsto \kappa_P(\Theta_\rho^* f), \qquad J_K = \operatorname{im} \eta_\rho^P.$$

This restricts a classical characteristic map; it is not a new construction of the ambient K-ring. The question is the arithmetic of the selected representation-dependent subring.

Theorem 10.1 (Finite-theory comparison; repository deduction)**.** *For the full literal source on a simple simply connected complex group, β^P and η_ρ^P have the same integral kernel. The induced isomorphism $J_H \simeq J_K$ is strict for polynomial degree on J_H and the induced support filtration on J_K, with*

$$\operatorname{gr} J_K = J_H \subseteq H^{2*}(G/P; \mathbb{Z}).$$

The filtration is the topological/support filtration, not an asserted equality with the representation-ring gamma filtration. Over $\mathbb{Q}$, positive-degree global invariants pull back to virtual G-characters of dimension zero and therefore vanish after the associated-bundle map. The first derivative $d\Theta_1 = \mathrm{id}$ makes the induced map triangular with identity leading term. The finite filtration gives an isomorphism over $\mathbb{Q}$; torsion-freeness of the ambient groups retains the kernel and leading-term statements integrally for the specified source.

This theorem does not identify the integral cokernels of the two embeddings. Filtered extensions can change Smith factors even when associated-graded lattices agree. It explains why ordinary cohomological results remain relevant while new K-theoretic arithmetic is possible. The total index in a full-rank setting is controlled by the associated graded; individual invariant factors require more information.

For example, the defining symplectic representation on $\mathrm{Sp}(2n)/P_1 \simeq \mathbb{P}^{2n-1}$ has

$$K = \mathbb{Z}[v]/v^{2n}, \qquad L = 1 + v, \qquad b = L - L^{-1}, \qquad J_K = \mathbb{Z}[b].$$

The literal generator is integral. Dividing it by two would give a rational class and change the question. Ordinary here means *nonequivariant K-theory*, not singular cohomology.

Theorem 10.2 (Symplectic projective Smith factors; repository deduction)**.** *Write $n = 2r + \epsilon$ with $\epsilon \in \{0, 1\}$, and let $s_2(a)$ count the ones in the binary expansion of a. For $0 \le j < r$, put*

$$a_{n,j} = \begin{cases} 4 + 8j + s_2(r - j - 1) - s_2(r + j), & n = 2r, \\ 8 + 8j + s_2(r - j - 1) - s_2(r + j + 1), & n = 2r + 1. \end{cases}$$

The inclusion $\mathbb{Z}[b] \subset K$ has n unit factors, one factor 2 if $\epsilon = 1$, and two factors $2^{a_{n,j}}$ for every j.

For $n = 1$ the list is $(1, 2)$; for $n = 2$ it is $(1, 1, 8, 8)$. The latter has the same total index as the cohomological diagonal $1, 2, 4, 8$, but a different Smith list. The proof reduces to a Hankel moment matrix, identifies its moments with a finite binomial continued fraction, and applies integral 2-local orthogonal-polynomial diagonalization. The continued fraction and Hankel method are published inputs, not new constructions of this program.

Beyond the lattice, the normalized first-order character class in the augmentation cotangent space supplies additional representation data for K-theory. The pair consisting of Λ_ρ and that normalized character class is sufficient for the embedded comparison developed in the repository. It is not an injective classification of representations or images. Tensor powers and direct sums test this distinction: matching a torus lattice alone need not imply matching an embedded K-image.

Status and sources. See the filtered comparison, the character-cotangent comparison, and the closed projective formula. Associated-bundle and equivariant characteristic-map foundations are classical; [14] is one reference. These particular image statements are internally checked deductions.

11 Equivariance and the specialization defect

Equivariant theories retain parameters for the torus action. In cohomology the coefficient ring is $H^*_T(\mathrm{pt}; \mathbb{Z}) = \mathrm{Sym}_{\mathbb{Z}} X$; in K-theory it is $R(T) = \mathbb{Z}[X]$. Forgetting equivariance sends additive parameters to zero or multiplicative characters to one. Although the ambient flag theory has well-controlled specialization, a selected image need not specialize faithfully.

If $J_T \subset A_T$ is the equivariant image and I_T the augmentation ideal of its coefficient ring, the relevant map is

$$J_T/I_T J_T \longrightarrow J_0 \subset A_T/I_T A_T.$$

Its kernel is

$$\Delta = (J_T \cap I_T A_T)/I_T J_T.$$

This measures the failure of forming the embedded image to commute with specialization. A source element can become zero in the specialized ambient ring without already being an augmentation multiple inside the source image. Rationalization can erase the defect entirely.

11.1 The defining type A projective family

Take $G = \mathrm{SL}_m$, $P = P_1$, and the defining representation. Put

$$R_T = \mathbb{Z}[t_1^{\pm 1}, \dots, t_m^{\pm 1}]/(\textstyle\prod t_i - 1), \quad A_T = R_T[L]/\prod_{i=1}^{m}(L - t_i), \quad I_T = (t_i - 1).$$

The line-class convention is fixed by this presentation. Define all jets

$$f_j = [u^{m-j}] \prod_{i=1}^{m}(L + u - t_i), \qquad J_T = R_T[f_1, \dots, f_{m-1}].$$

These are the images of the full literal root-polynomial Levi source. The identity $\Theta^*(x_i - x_j) = t_i - t_j$ connects the jets directly to the original Springer map. Let Δ_m be the kernel of specialization to its ordinary K-image.

Theorem 11.1 (Uniform specialization bound and sharp exponent)**.** *For every $m \geq 2$, $m\Delta_m = 0$. Specialization is faithful exactly when $m = 2$. For every $m \geq 3$, Δ_m has exponent exactly m, and its nonzero primary parts occur at precisely the primes dividing m.*

The assertion is about the full multiplicative image, including products whose total degree exceeds the defining polynomial degree. It is not merely a bound on a span of jets or on a low-degree truncation. In particular $m\Delta_m = 0$ says that the extra specialized classes are m-torsion; it does not say that $m(A_T/J_T) = 0$.

The proof is a general coefficient-family argument. For a monic $F(X)$ of degree m over an augmented $\mathbb{Z}$-algebra S, suppose F specializes to $(X-a)^m$. Define $A = S[L]/F(L)$ and let J be generated by every nontrivial jet of $F(L+u)$. Integral difference traces and Hasse translation construct witnesses for

$$m(J \cap IA) \subseteq IJ.$$

The recurrence uses the monic relation $F(L) = 0$ to handle overflow products uniformly. Universal integral identities can then be evaluated in the determinant-one coefficient family without assuming that images commute with base change. The sharper lower bound uses explicit torsion detectors; a separate 2-primary argument is needed when $m = 2 \pmod 4$.

The full-flag transfer gives, ungraded, $(m-1)!$ copies of the projective defect, with graded shifts retained in the precise statement. For a parabolic with Levi block sizes n_i, the same free-source transfer works locally at primes not dividing $\gcd(n_i)$, and integrally when that gcd is one. An $m\gcd(n_i)$ annihilator holds in the general recorded parabolic scope, but sharpness at the excluded primes is not proved.

11.2 What equivariant structure adds

Over a multivariable coefficient ring, a single Smith list is generally not the correct invariant. Presentation matrices and their Fitting ideals record rank drops and torsion support under specialization. The symplectic projective work produces doubled-block and Hankel models, integral Schur generators, and balanced-product formulas for all Fitting ideals in that family. These are stronger than a determinant alone, but are not a classification of primary decompositions after arbitrary substitution.

A cohomological negative control is the literal defining SL_4/P_1 source: specialization already has a $\mathbb{Z}/2$ kernel in polynomial degree five. It must not be conflated with Δ_m above, which lives in K-theory. Both examples show that the apparent simplicity of the ambient flag theory does not force faithful specialization of the selected representation image.

Status and sources. See the actual source and faithfulness theorem, the uniform recurrence, equivariant methods, and the cohomological control. The recurrence and exact exponent are repository proofs, not consequences of a finite rank survey.

12 Partition multiplicities and the limit of the present K-theory theorem

The exponent theorem does not determine how many cyclic summands occur. The refined problem asks for the complete primary multiplicities of Δ_m in the grading transported from its universal homogeneous model. Let $P_k(n)$ count partitions of n into parts at most k, with $P_k(0) = 1$ and $P_k(n) = 0$ for negative n. Define

$$a_d = \sum_{k\geq 2} P_k(d-(2^k-1)), \qquad \sum_{d\geq 0} a_d q^d = \sum_{k\geq 2} \frac{q^{2^k-1}}{\prod_{i=1}^{k}(1-q^i)}.$$

The initial values are $0,0,0,1,1,2,2,4,4,6,7$ for $d = 0,\dots,10$.

Theorem 12.1 (Established primary ranges; repository deductions)**.** *Let $p \mid m$ and $s = v_p(m)$. For $d \le p$,*

$$(\Delta_m)_{d,(p)} \simeq (\mathbb{Z}/p^s)^{a_d}.$$

The same formula holds for $d = p+1$ when p is odd. At $d = p+2$, for odd p and $m > p$, it becomes

$$(\Delta_m)_{p+2,(p)} \simeq (\mathbb{Z}/p^s)^{a_{p+2}-1} \oplus \mathbb{Z}/p^{s-1}.$$

At $d = p+3$, for $p \ge 5$ and $m > p$, the uncorrected formula returns:

$$(\Delta_m)_{p+3,(p)} \simeq (\mathbb{Z}/p^s)^{a_{p+3}}.$$

At the distinct boundary $m = p$, the last two established degrees instead have $(\mathbb{Z}/p)^{a_d-2}$, with the same respective restrictions on p. A factor of order p^0 is trivial.

The prime boundary and the root boundary are different. The prime boundary is the passage from degrees below p to degrees where Newton/Witt coordinate changes involve p. The root boundary $m = p$ additionally changes the coefficient family and its root equation. Treating it as a generic $m > p$ case would give the wrong answer.

The stable partition formula arises from a universal moment source and dual constraints indexed by subset hyperplanes. Integral saturation is proved in the initial range, including a divided-power argument at $d = p$. The first Witt corrections track both changes to the source and changes to the coefficient ideal. At $p+2$, smaller-order torsion appears; at $p+3$, a common ambient torsion block cancels relatively, while an added Witt source direction must still be retained. The theorem therefore depends on exact image calculations, not just the Hilbert function of a rational moment algebra.

There is a useful all-degree recurrence for high coefficient degrees after inverting the relevant excess factorial. It settles the nonpolarizable sectors needed for the proposed first Frobenius band $p \le d < 2p$ at odd primes. What remains is the uniform low-coefficient tensor-obstruction comparison, followed by the full elementary Witt comparison and boundary analysis. The complete band is not proved. Higher 2-primary pieces are especially important because arguments dividing by two cannot address them.

An explicit order-five relative obstruction in coefficient degree three and weight twelve disproves a blanket relative-saturation principle. This is a constraint on the proof strategy, not an inconvenience to remove by localization. A successful general theorem must explain when such torsion cancels in the actual specialization defect and when it survives.

The research value of this direction is a representation-dependent base-change theorem with sharp integral failure terms. The ambient K-theory and the combinatorics of partitions are established subjects. The target contribution is the exact arithmetic of their interaction through the Springer source. A credible generalization must retain the full image, rather than a convenient linear approximation to it.

Status and sources. The chain of proofs is the stable range, the first correction, the moment obstruction, and the next elementary layer. Historical paragraphs in earlier notes are superseded only in these explicitly completed degrees.

13 Oriented, quantum, and affine extensions

The finite ordinary and equivariant work supplies several entry points to other theories. The objective is not to rebuild their ambient Schubert calculus, which is already extensively developed. It is to specify and analyze the representation-dependent image within it.

13.1 Oriented theories

An oriented theory has a formal group law controlling first Chern classes of tensor products. Additive cohomology and multiplicative K-theory are different specializations. Formal Demazure and formal group-ring frameworks provide published characteristic maps under their stated coefficient and torsion hypotheses [15, 16]. Our finite-theory notes define a specified universal lift and compare its images under specialization. Torsion-free comparisons do not imply modular injectivity: additional kernels can be measured by a Tor_1 term.

The existing type A and symplectic projective calculations provide controlled height and Fitting-ideal benchmarks. A broader project should begin with a precise coefficient ring, formal group law, and source, then ask which integral image data survive. It should not assume that the torus degree-one theorem for the additive trace construction automatically classifies every possible formal-group-law lift. Higher-generator Levi sources introduce relation-lifting questions absent in a one-generator pilot.

13.2 Quantum images

On the full flag, a polynomial source on linear generators admits a specified divisor lift using quantum multiplication. On a partial flag, higher generators and relations require separate justification. The old ordinary kernel generally changes: on $\mathbb{P}^1$, the image of a square of a divisor is a nonzero quantum parameter.

For the defining SL_m/P_1 literal source, the recorded lift sends its elementary generators to $\binom{m}{j}h^j$ in $\mathbb{Z}[q,h]/(h^m-q)$. Put $g_0=1$ and $g_r=m/\gcd(m,r)$ for $0<r<m$. The complete image is

$$J_q=\bigoplus_{r=0}^{m-1} g_r\mathbb{Z}[q]h^r.$$

It specializes faithfully at $q=0$. With $y_r=g_rh^r$, multiplication is

$$y_ay_b=\frac{g_ag_b}{g_{(a+b)\bmod m}}q^{\lfloor (a+b)/m\rfloor}y_{(a+b)\bmod m}.$$

Primewise divisibility of the coefficients proves closure. This is an elementary but complete image benchmark, not a quantum theory for all parabolics. The symplectic projective family supplies a second specified benchmark. More interesting partial flags require a canonical lift or an explicit explanation of the choices, and then a proof of flatness or a description of its failure.

13.3 Two different affine routes

The Peterson correspondence relates localized quantum cohomology to a quotient of affine Grassmannian *homology*, with Pontryagin product [17]. Transporting a specified quantum image through that correspondence is meaningful, but recovering an unlocalized integral image requires additional work. Localization can erase precisely the arithmetic under study.

A separate ordinary cup-cohomology construction uses evaluation and level. For a compact form K, the topological affine flag LK/T identifies with $\Omega K\times K/T$. Pull back the finite image by evaluation and adjoin a chosen multiple $k\eta$ of the loop-space degree-two class. The level k is additional data, not automatically determined by the normalized Springer map. In type SL_n, if $c_{n,d}$

is the content of η^d, then

$$v_p(c_{n,d}) = v_p\left(\left\lfloor \frac{d}{p^{a_p}} \right\rfloor !\right), \qquad a_p = \max\{a \geq 0 : p^a < n\}.$$

This follows by optimizing the valuations of multinomial coefficients in the integral symmetric-function Hopf model. The content of $(k\eta)^d$ is $k^d c_{n,d}$. It describes a selected affine image and does not classify the full affine flag ring. Published evaluation/divisor formulas are used in fixing the convention [18].

Pointwise application of a finite Springer map already gives a map on loops, and diagram-compatible cases give twisted versions. A naive descent to an affine flag quotient fails: right multiplication by an Iwahori element is not conjugation equivariance. Central and rotation directions introduce further questions. Thus a raw loop map, an affine-flag characteristic image, and an affine Kac–Moody trace-projection construction are three different tasks.

Status and sources. See oriented comparisons, height benchmarks, quantum images, and the affine construction. Published ambient theories are cited above; broader image classifications and descent statements remain research directions.

14 The generalized Cayley side: information before the Borel quotient

The trace-projection map in this program is the generalized Cayley map studied by Kostant–Michor. Their work relates its degree to the induced map on adjoint quotients and uses its differential in a harmonic-separation problem [5]. The geometric branch of the present program retains positive-degree global invariant functions, which the Borel map kills. It therefore studies additional information in the same original representation-dependent morphism.

14.1 Quotient coordinates and the full differential

For basic invariants $f_1, \ldots, f_r \in \mathbb{C}[\mathfrak{t}]^W$, the functions $f_i(\Theta_\rho(t))$ describe $T/W \to \mathfrak{t}/W$. Fundamental characters give affine coordinates on T/W for a simply connected semisimple group [7]. The exceptional API implements exact higher coordinates and checks invariant degrees and independence, rather than identifying a few low-degree expressions with a complete invariant system.

In logarithmic torus coordinates, the derivative matrix is

$$J_T(t) = M_\rho^{-1} \sum_\mu m_\mu \mu \mu^t e^\mu(t).$$

This is only the Cartan block. Equivariance determines each root-direction eigenvalue, yielding the full left-trivialized Cayley determinant

$$\Psi_\rho|_T = \det J_T \prod_{\alpha>0} e^\alpha \left(\frac{\alpha(\Theta_\rho)}{e^\alpha - 1}\right)^2.$$

The apparent quotients extend as Laurent polynomials, because they are entries of the regular differential. The formula initially follows on a dense torus open and then extends. It is a direct deduction from the explicit torus map and equivariance. Its zero set is a *source* critical locus; it is not already a computation of the target branch or nonproperness locus.

The same equivariance permits reconstruction of Borel-coordinate formulas from torus data and root commutators. Character-based recurrences extend the initial SL_3 and Sp_4 calculations to other representations and root data. Their value is a uniform mechanism for root coordinates and collapse tests. Singular root coefficients mark places where unipotent directions can cease to be determined by first-order data; those loci require the higher recurrence, not a generic division by a root coefficient.

14.2 Generic degrees and boundary corrections

For a dominant generically finite map, degree means the characteristic-zero extension degree of rational-function fields. In this setting

$$\deg \theta_\rho = \deg(T \to \mathfrak{t}) = \deg(T/W \to \mathfrak{t}/W).$$

The two Weyl quotient degrees cancel; one does not divide the torus degree by $|W|$. For an adjoint character the Newton calculation often takes place on the root-lattice torus, so a central covering factor must be restored for the simply connected group.

The recorded exceptional calculations are:

Representation	Volume	Boundary	Cover	Degree
E6 minimal, 27	1080	0	1	1080
E6 adjoint, 78	2808	1890	3	2754
E7 minimal, 56	9792	2520	1	7272
E7 adjoint, 133	21312	4480	2	33664

The degree is (volume − boundary) × cover. A raw Newton volume is wrong in three of these cases. The E6 correction 1890 also has a deformation reconstruction, supplying a different check on the intersection calculation. F4 minimal degree 96, F4 minimal/adjoint mixtures, and defining/adjoint mixtures in B/C have separate recorded boundary formulas and exceptional coefficient ratios. These extend the method to coherent families, but do not constitute an arbitrary-representation boundary algorithm.

The unresolved structural issue is how to derive all contributing face schemes and their normal multiplicities from representation data. A facet character alone may not determine the correction. Toric intersection theory is the published foundation; the representation-specific boundary calculations are the additional work. These degrees are degrees of particular regular maps, not the minimum Cayley degree over all dominant rational equivariant maps. Historical minimum-degree questions therefore cannot be declared solved by this table.

Status and sources. See quotients and differential, exceptional degrees, boundary criteria, F4 mixtures, and B/C mixtures. Exact certificate checks support the stated cases; external review and literature priority remain separate.

15 Regularity preservation and harmonic separation

An element is regular if its centralizer has dimension $\operatorname{rank} G$. Torus regularity means regular semisimple points of T map to regular points of $\mathfrak{t}$. Global regularity preservation also includes regular nonsemisimple group elements. These are distinct conditions, and neither is the same as generic smoothness of a specified fiber family.

The repository has a uniform classification for actual nontrivial finite-dimensional representations of simple simply connected complex groups, allowing direct sums and trivial summands. For SL_N, $N \geq 3$, the globally preserving representations are copies of the defining module, or copies of its dual, with optional trivial summands; mixing the two nontrivial types is excluded. For SL_2, only copies of the defining module plus trivial summands preserve global regularity. No global positive cases occur in the remaining simple types under these hypotheses.

There is a larger torus-preserving symplectic family. Write $D(g) = \det(I + g)$ in the defining symplectic module. Actual characters of the form $AD^m + B$, with A, m positive integers and B chosen so the expression is an actual character, preserve torus regularity. In rank at least two they fail global preservation. The same torus family occurs for $\mathrm{SL}_2 = \mathrm{Sp}_2$, but global preservation is more restrictive. The classification does not cover arbitrary virtual characters with negative nontrivial multiplicities, positive characteristic, or general reductive products.

The argument exploits positivity of actual weight multiplicities on real root directions, Laurent factorization of root-coordinate functions, and compatibility across the root system. Positivity is a hypothesis with content, not a property of arbitrary invariant potentials. The standard defining SL_N example is already treated by Kostant–Michor; the broader classification is a working deduction and requires focused external review.

15.1 A sharp localization that does not require global regularity

Let $G = \mathrm{Sp}(2n)$, $n \geq 2$, and let θ_m be the normalized map for a determinant-power character. Put $A_G = \mathbb{C}[G]$, $R_G = A_G^G$, and $\mathcal{H}_m = \theta_m^* \operatorname{Harm}(\mathfrak{g})$. The harmonic space is a vector space, not a subalgebra. The recorded theorem is

$$R_G[D^{-1}] \otimes_{\mathbb{C}} \mathcal{H}_m \xrightarrow{\sim} A_G[D^{-1}].$$

Among nonzero invariant principal localizations by f, this multiplication map is an isomorphism exactly when D divides f. Thus the determinant divisor is the minimal support to remove within that specified class of localizations. The full normalized Jacobian is

$$\Psi_m(g) = \left(\frac{D(g)}{4^n}\right)^{(nm-1)(2n+1)} \left(1 - nm + \frac{m}{2}\operatorname{tr}(g)\right).$$

Separation holds across the extra critical divisor in the last factor. This refines the needed localization for this fixed pulled-back harmonic space; it does not contradict the global regularity obstruction or remove poles from every inverse-differential construction. The proof uses the classical Cayley isomorphism on $D \neq 0$ and known Lie-algebra harmonic separation.

Status and sources. See the complete classification, the sharp localization, and the literature audit. These results concern the generalized Cayley map itself. They do not propose an unrelated replacement harmonic complement.

16 Actual adjoint fibers: the completed case theorem

Let G be simple and simply connected, $\mathfrak{g} = \mathrm{Lie}(G)$, and B_{Kill} its Killing form. Define

$$B_{\mathrm{Kill}}(\theta_{\mathrm{Ad}}(g), X) = \operatorname{tr}(\mathrm{Ad}(g) \operatorname{ad} X).$$

For a proper Levi subgroup K with Lie algebra $\mathfrak{k}$, put

$$Z_K^\circ = \{h \in \mathfrak{z}(\mathfrak{k}) : Z_G(h) = K\}.$$

Theorem 16.1 (Generic adjoint regular-source fibers; completed repository case proof)**.** *For every such G, K, there is a dense Zariski open $U_K \subset Z_K^\circ$ such that the actual scheme*

$$\theta_{\mathrm{Ad}}^{-1}(h) \cap G_{\mathrm{reg}}$$

is smooth for every $h \in U_K$. Empty fibers and smooth positive-dimensional transverse factors are allowed. After further shrinking, this is a smooth family over U_K.

The statement includes all finite simple types, including E8. It retains nilpotents in the equations and regular nonsemisimple source points. It does not replace a fiber by its reduced support. It does not assert that every centralizer-exact target is good, or that the fiber is nonempty, or that its dimension always equals $\dim K - \operatorname{rank} G$.

16.1 The common scheme reduction

Equivariance gives $(\mathrm{Ad}(g) - 1)\theta(g) = 0$. On the fiber over h, this places the centralizer equations in the fiber ideal, so the actual fiber lies in K. On a finite central etale cover $[K, K] \times Z(K)^\circ \to K$, use fundamental characters $x_1, \ldots, x_\ell$ of the derived group and central torus coordinates $z_1, \ldots, z_s$. Let $\Phi(x, z)$ be the *full* restricted adjoint character. Define

$$Y = V(\partial_{x_1}\Phi, \ldots, \partial_{x_\ell}\Phi), \qquad \mu_j = z_j \partial_{z_j}\Phi.$$

The transverse scheme over a target c is cut out by these equations and $\mu_j = c_j$. On the regular source, the adjoint quotient is smooth, and the completed local ring of the group fiber is the completed transverse ring with $\dim K - \operatorname{rank} G$ formal smooth variables adjoined. The finite central cover is etale; later Weyl trace substitutions can ramify and require separate treatment.

The uniform obstruction is therefore

$$\overline{\mu(\operatorname{Sing} Y)} \subsetneq \mathfrak{z}(\mathfrak{k}),$$

where $\operatorname{Sing} Y$ means the nonsmooth locus of the actual scheme. Standard generic smoothness applies only after this image is excluded. It cannot remove a persistent nonreduced component that dominates the target.

16.2 What a certificate means in this proof

For exceptional cases, a certificate is finite exact algebraic evidence for a precisely stated step: original polynomial generators, ideal membership witnesses or certified leading terms, dimensions or degrees, and the hypotheses linking those data to a characteristic-zero statement. It is not a numerical sample or a second name for the theorem. A monomial dimension count checks an initial ideal once that initial ideal has been justified. A finite-prime computation needs a valid characteristic-zero implication; it does not supply one by itself.

The consolidated proof handles all 255 proper E8 standard subsets through Weyl transfers and the remaining critical models. A fresh audit rechecked transfers, original critical generators, and the passage to actual group fibers. The last degree-at-infinity argument has an explicit finite-prime membership dependency on a pinned exact engine. Native checks independently verify specified identities and monomial counts, but do not independently reproduce every engine membership. That limitation should be disclosed to a reviewer rather than hidden by the phrase “independently checked.”

Three controls remain indispensable. A special Sp_4 centralizer-exact target has a singular transverse fiber, so genericity is essential. An E6 contact example has transverse excess two, so universal quasifiniteness is false. Altered characters and an actual nonadjoint SL_3 representation give persistent nilpotent behavior, so Weyl symmetry and positivity alone do not prove the adjoint theorem. The case proof is complete in its recorded scope; a uniform root-theoretic replacement is still unresolved.

Status and sources. Precise theorem and dependencies: the consolidation; the fresh E8 audit; the review package. This is a computer-assisted working proof, not external mathematical certification.

17 Nonemptiness, excess, and representation changes

Smoothness allowing empty fibers leaves an important classification problem: which components actually dominate a central target stratum, and how large are their fibers? The repository derives a critical-value/stabilizer criterion and a rational-linear description of the relevant central images. These are structural tools, but finding all components directly from roots remains unresolved in general.

For actual representation characters polynomial in the adjoint character, write $\chi_\sigma = f(\chi_{\mathrm{Ad}})$ and assume the trace normalization is nondegenerate. The chain rule gives

$$\theta_\sigma = \frac{f'(\chi_{\mathrm{Ad}})}{f'(\dim G)}\theta_{\mathrm{Ad}}.$$

On a fiber over a nonzero central target the numerator scalar is a unit. On a dominant component with constant adjoint-character value b, the component survives exactly when $f'(b) \neq 0$. Smoothness and excess are then transported by the proved component criterion. This is not a classification of arbitrary irreducible representations; actual polynomial characters are a specified family.

17.1 Classical block classification

For SL_N with proper Levi block sizes b_i, generic nonemptiness in this polynomial-adjoint family occurs exactly when every $b_i \leq 2$ and at least one block is a singleton. The dimension is $\sum_i b_i(b_i-1)$. In B/C/D, write the matrix Levi pattern as $\prod_i \mathrm{GL}_{b_i} \times H_m$, with $\sum b_i + m = n$ and residual H_m orthogonal or symplectic. Absorb SO_2 into a GL_1 block in D.

If some $b_i > 4$, or if $m > 2$ in B, $m > 1$ in C, or $m > 3$ in D, the regular fiber is empty over centralizer-exact targets. Within those bounds, all patterns other than the all-four-block families and two small residual exceptions have a dominant component with nonconstant adjoint character, and therefore remain generically nonempty for every admissible f. The exceptional patterns have explicit finite constant-value sets; the criterion is that f' not vanish on all those values. The two isolated patterns are B3 with $\mathrm{GL}_1 \times \mathrm{SO}_5$ and D4 with $\mathrm{GL}_1 \times \mathrm{SO}_6$, both governed by $f'(0)$.

Whenever nonempty, the generic fibers are smooth of dimension

$$\sum_i b_i(b_i-1) + 2m^2 \quad \text{in B/C}, \qquad \sum_i b_i(b_i-1) + 2m(m-1) \quad \text{in D}.$$

Thus the transverse excess is zero throughout this classical family. On GL_4^q with no residual factor, the constant value is $-q/(2q-1)$ in B and zero in C/D. In particular pure adjoint tensor powers of order at least two erase the C/D strata. The exceptional E6 contact stratum behaves differently and demonstrates why this classical zero-excess conclusion cannot be promoted to every simple group.

There is also a recorded extension of adjoint generic smoothness to semisimple groups and all Levi factors, using a separate regular zero-fiber calculation and finite central descent. It is not an automatic reductive extension, since the adjoint trace form is degenerate on a central torus. The present exposition keeps the proper-Levi simple theorem as its principal statement because that is the scope of the original conjecture and its review package.

Status and sources. See the structural criterion and zero targets, polynomial-character survival, and the full classical table and boundary audit. The latter table, rather than an inferred extrapolation from the examples here, gives the complete constant-value tests.

18 A prioritized research program

The immediate objective is a structural account of integral Springer–Borel images. The Cayley results form a substantial related project, but reopening a long exceptional census would not advance the current ordinary-image question. The following priorities separate an active proof problem, consolidation needed for publication, and longer-term extensions.

18.1 Priority one: a short presentation of classical saturations

Target. Prove or refute $J_e = K_e$ for $\ell \geq 3$ and $1 \leq e < \ell$ in the square-zero defect model. Together with the proved denominator formula, this would turn the complete B/D square Smith theory into a uniform small-generator description of the saturated ring.

Route. Work with the coefficient images of $\bigwedge^{\ell}(1 + \sum_{j \geq c} a_j T^j)$ over $\mathbb{Z}$. Seek a straightening order with monic reductions and a basis whose degree distribution matches the defect variety. The two-variable proof supplies the correct standard: integral freeness plus the known rank, not rational rank alone. An alternative route is an integral incidence filtration that identifies these operator images with primitive geometric subquotients.

Decisive test. The argument must retain all coefficient operators, divided-power multiplicities, and boundary terms. A proof after inverting factorials is partial. A counterexample must identify a specific integral class in K_e/J_e and explain whether the failure is rank or torsion. Either outcome would be progress: the former proves the proposed presentation; the latter identifies the missing generator mechanism.

Literature comparison. Compare the exact Grassmannian quotient and integral coefficient ring with truncated symmetric ideals, Tanisaki-type presentations, and integral current-algebra modules. Similar Hilbert series are useful clues but not identifications. In particular stable-plane conditions and annihilated-flag conditions differ, and characteristic-zero module bases cannot be imported as integral primitivity theorems without a compatible lattice map.

18.2 Priority two: consolidate the ordinary theorem package

Target. Prepare the source-lattice theorem, B/D Smith formulas, adapted bases, and geometric denominator theorem for expert review as a coherent package. A review should separate published integral Schubert inputs from the additional primitive-prefix and gluing arguments.

Route. Supply a minimal dependency graph, fix a single notation dictionary, and expose the key integral assertions as standalone lemmas. The full D gluing step and the use of Chow envelopes deserve particular attention. Replace any computationally observed statement with a proof or restrict

its claim to the certified cases. The new statement is a guide to this work, not a substitute for that review.

Success. A reader can verify the theorem without reconstructing a conversation or traversing historical checkpoints. The statement should explain exactly how the new finite integral results refine Parts II and III, while identifying stable injectivity and the underlying Chern expressions as published inputs. A separate novelty comparison should identify the exact integral-image formulation, rather than claim novelty for the Gaussian counting polynomial or standard straightening machinery.

18.3 Priority three: complete a coherent specialization range

Target. Determine the complete type A projective K-specialization defect in the first Frobenius band $p \leq d < 2p$, with exact boundary rules and a separate plan for $p = 2$.

Route. Use the proved high-coefficient recurrence to remove the settled sectors. Concentrate on the remaining low-coefficient binary-hyperplane tensor comparison, then track the actual elementary source through the required Witt changes. The order-five obstruction must be accounted for explicitly; a universal relative-saturation lemma is no longer a viable shortcut.

Success. A formula valid for all m in its stated prime and degree range, with witnesses and independent torsion detectors. A further table of ranks would not meet this criterion. If a full band is too ambitious, a theorem characterizing which obstruction modules control its factors would still be a meaningful general advance.

18.4 Priority four: structural exceptional and multiblock arithmetic

Two worthwhile questions remain distinct. The first is a root-theoretic explanation of the complete E6/E7 relative formulas: why the observed exponents and first-gap degrees occur, and when an analogous pattern holds for other root data. The second is the ambient quotient H/N, especially where root and weight lattices agree. A theorem about one does not solve the other.

For classical multiblock flags, the natural target is the complete ambient B/D joint-image filtration. Start from the established primitive separate-image sum inside the joint algebra, then identify the missing ambient mixed-product lattice. The right proof should be compatible with many blocks or explain the exact obstruction to such compatibility. It should not simply re-run the single-tail Smith routine for a longer list of examples.

18.5 Parallel review and longer-term geometry

The completed adjoint fiber theorem should undergo focused external review before a uniform replacement proof is made the dominant project. The review has concrete tasks: audit the scheme-preserving reduction, the E8 characteristic-zero certificate implication, and the passage through central covers. Independent review means a mathematician's assessment of the argument, not merely a second run of the same software.

A future uniform proof needs an additional identity forcing the singular image to be nondominant. The radial Casimir identity, finite central-image statements, and Hessian symmetry alone have not supplied it. Positive excess in E6 and persistent nilpotents for altered characters are mandatory controls. A useful new lemma must imply a scheme-theoretic conclusion beyond those already-known constraints.

Degree questions should similarly be organized around an effective boundary theorem for a family of characters, with the correct Newton lattice and all normal multiplicities. Minimum Cayley degree, target branching, and nonproperness are separate projects. Historical claims of openness should be checked anew before becoming a proposal's central motivation.

19 Connections, limitations, and standards of evidence

19.1 How the work extends the papers

The connection is direct at the level of maps. The source is obtained using the same Θ_ρ^*, the cohomological map is the same ξ_ρ^P, and the normalized classical generators are those of the published calculations. The extension changes the questions asked of these maps:

1. The explicit weight formula becomes a general arithmetic description of the full integral torus source.
2. Complex or stable injectivity becomes a finite-rank study of kernels, saturation, and every Smith multiplicity for selected classical sources.
3. Abstract divisibility becomes explicit divided classes and integral multiplication in the saturation.
4. The same integral polynomial characters enter K-theory, where support filtrations and equivariant specialization reveal additional arithmetic.
5. Before the Borel quotient, the same trace map supplies quotient coordinates, differential factors, degree corrections, and actual-fiber equations.

The ambient Borel, BGG, Schubert, characteristic-map, and harmonic theories remain published foundations. This program does not claim to have introduced them. The candidate contributions lie in the precise representation-dependent image and fiber statements, subject to the proof and novelty qualifications already stated.

19.2 Hiller, folding, and adjacent geometry

Hiller's chain/intersection approach suggests ways to interpret products and incidence filtrations in Schubert calculus [9]. A comparison with the Smith prefixes would need an integral map and a primitivity theorem. Matching a chain count to a multiplicity polynomial would not establish the integral identification.

Diagram folding gives a different connection. For a diagram-stable representation, trace-orthogonal projection is compatible with restriction to the connected fixed subgroup. The full integral source can nevertheless enlarge after restriction. The recorded SL/Sp, E6/F4, and Spin8/G2 comparisons exhibit this defect. A folding explanation of an E7 multiplicity expression therefore requires more than noticing a smaller-type Weyl polynomial: it must identify the image lattice and account for its index.

The square-zero varieties suggest Springer-fiber and Spaltenstein geometry, but the incidence conditions must be matched exactly. This is a promising bridge between classical representation combinatorics and the denominator filtration. The most useful next connection would prove an integral restriction-kernel statement, not merely provide another interpretation of the same rational generating function.

19.3 What remains unproved

The principal unresolved claims are: the higher-block intermediate kernel equality; full type A specialization multiplicities beyond the stated ranges; ambient B/D joint-image factors; general ambient exceptional image formulas; a uniform root explanation of the completed adjoint case theorem; and broad exceptional nonemptiness/excess classifications. General direct-sum lattice formulas, bad-prime invariant sources, and arbitrary-representation boundary degrees remain further structural questions.

Several stronger statements have already failed and should not reappear as tacit assumptions. Images need not commute faithfully with specialization. The torus lattice alone need not determine the K-image. Universal zero transverse excess is false. Global regularity preservation is much rarer than generic fiber smoothness. A naive root-power extension of the square-generator ideal is false. Rational Hilbert-series equality does not prove integral saturation, and a reduced-support calculation does not detect embedded nilpotents.

19.4 Verification policy

Each claimed theorem should have a written argument, a source attribution, and a sharply delimited computational dependency. Independent checks are strongest when they change the mathematical route: BGG versus Chevalley for Schubert coefficients; symmetric-function versus native integer straightening for lattices; companion matrices versus Newton identities for character formulas. Agreement between two wrappers around the same engine is not independent verification of that engine.

The latest ordinary package records 96 B embeddings, 124 constructed D components, 28 actual D embeddings, and seven D product checks; separate native Chevalley calculations cover 26 B and 28 D components. The two-variable audit verifies monic integral reductions, including 283 S-pairs over fifteen parameter choices. These checks detect implementation and transcription errors. The all-rank claims rest on the written filtration and recurrence proofs. The statement's rendering and link checks are a separate editorial audit and do not strengthen the mathematical proof status.

Status and sources. The broader framework and exact failed generalizations are indexed in the ledger. Folding is treated in the restriction note. The release record binds the latest ordinary checks to their scripts and proof inputs.

A Result inventory and review guide

This inventory records the principal results, rather than every historical lemma or finite calculation. "Proved" in the status column means a written repository proof with the stated internal checks, unless the row says published. It does not establish literature priority. The linked baseline fixes what a collaborator is being asked to review.

Record	Result and scope	Status and remaining qualification
BASE-02/03	Explicit Springer formula and polynomial-character Borel maps	Published Rogers/Kumar framework; Parts II/III supply stable and explicit families.
SRC-01/03	Full torus source is $\mathrm{Sym}_{\mathbb{Z}} \Lambda_\rho$; Levi source is its invariant ring	Proved in the trace-nondegenerate simply connected simple scope; Levi invariants need not be generated in degree one.

Record	Result and scope	Status and remaining qualification
SRC-02	$\Lambda_\lambda = I_\lambda(Q + \mathbb{Z}\lambda)^*$	Proved for every nontrivial irreducible finite simple type; direct sums require combined coefficient lattices.
IMG-10/11	Torsion-index bound and exact defining projective factors	Proved full literal source statements; general bad-prime multiplicities not determined.
IMG-02–05	C primitivity; B factors; D square equality and full gluing	Proved all-rank normalized single-tail results; literal scaling must be specified separately.
IMG-08/09	C joint primitivity and B/D relative sum theorem	Proved for specified two-step generators; ambient B/D joint factors remain open.
IMG-12/20	E6 all 63 and E7 all 127 proper relative root/weight comparisons	Complete, partly computer-assisted; not complete ambient exceptional images.
IMG-44/45	Adapted B/D bases and full D multiplication	Proved constructions from previous prefix/gluing theorems; independently checked finite embeddings.
IMG-46/47	Geometric denominator formula; top layer; all layers for $\ell = 2$	Proved; $J_e = K_e$ for intermediate e and $\ell \geq 3$ remains conjectural.
IMG-27/31	Filtered finite K-image comparison; symplectic projective Smith list	Proved in the specified literal-source scope; not a lattice-only K-image classification.
IMG-37/39	Type A K-specialization faithfulness and exact exponent	Proved all m: $\Delta_2 = 0$, exponent m for $m \geq 3$.
IMG-40–43	Partition factors and corrections through $p + 3$	Proved only in the prime/degree ranges stated in Section 12; full band and higher 2-primary pieces open.
IMG-32/33	Specified projective quantum and evaluation/level affine images	Proved benchmark constructions; arbitrary partial-flag lifts and affine descent remain separate.
GEO-05 and degree notes	E6/E7 minimal/adjoint generic degrees and further families	Exact working deductions with boundary dependencies; not minimum Cayley degrees.
GEO-12/13	Symplectic sharp harmonic localization and regularity classification	Written internally audited deductions; external review recommended.
GEO-90–92	All-simple proper-Levi adjoint generic regular-source smoothness	Completed by cases; E8 certificate has an explicit engine dependency; uniform explanation open.
GEO-94–97	Semisimple extension; polynomial-character survival; classical blocks	Proved with exact hypotheses; arbitrary representations and exceptional excess classification open.

Suggested reading paths

For a collaborator primarily interested in Parts II and III, read Sections 2–9, then the B/D proof notes and the two-variable straightening proof. For a collaborator interested in specialization, read Sections 4–5 and 10–13 before the recurrence and Witt-comparison notes. For a geometric reviewer of the adjoint theorem, read Sections 14–17 and then the isolated review package, keeping the common scheme reduction distinct from the exceptional certificates.

An initial external review should focus on either the primitive-prefix/gluing chain or the actual-fiber/E8 chain. It can identify gaps, simplify arguments, or recognize prior results without endorsing the entire program.

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Scope of the bibliography. These primary references locate the construction and the principal adjacent theories. Links and bibliographic identities were checked during preparation; the statement also uses the detailed source comparisons already recorded in the pinned proof notes. This is not a new exhaustive citation census or a proof that any working result is absent from the literature. The repository links throughout the text supply the exact additional arguments, computations, and limitations being proposed for review.